UM-4 provenance

Where every input of UM-4 comes from, every decision taken, every computation and its checks, the external data and software, and how to reproduce all of it

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Provenance of UM-4, 2026-10-25

Repository once-ere/Dirac_claude, folder universe_model_4

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1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \ldots, x_7) :\ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure number Λ_{N} in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_{\mathrm{N}} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).
- Λ_{N}: The author's pure number in the vacuum equations of his notebook, $\Lambda = -\Lambda_{\mathrm{N}} H^2$: his original value 10; 21 in UM-4 (Einstein gravity), 7 in UM-3. Unit: dimensionless. Value: 21 (the author's original 10). Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)} = 1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)} = w_{(3)} = 0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda = -161H^2/16$); UM-3 used $(1, 1/20, 1/360)$. Unit: dimensionless. Value: $(1, 0, 0)$ in UM-4; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The metric of UM-4 depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one surface of $\mathcal{F}$, at $z = \pi/2$: $\mathcal{F}$ is the union of these copies, one per period (example 2), and "the fold" without a qualifier is the copy in the period that contains our world. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on

the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.

- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4. Notation: ln is the natural logarithm, e Euler's number, $\exp(y) = \mathrm{e}^y$, sin, cos and $\cot = \cos / \sin$ the trigonometric functions, sinh and tanh the hyperbolic functions, $|y|$ the absolute value; $f(0^+)$ and $f(0^-)$ are the limits of $f(\sigma)$ as σ approaches 0 from above and from below.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln \sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the metric of UM-4 is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] \tag{FD-4}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. The function $\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|}$, which multiplies the squared steps of all seven directions of the fold (ordinary space, the time and the extra times), is the warp factor of the metric (its square root $\mathrm{e}^{-H|\sigma|}$ multiplies the lengths). The time x_4 is warped like the other directions (the author's order of 2026-10-25, 13:15 PDT; in UM-3 it was not), so that every surface $\sigma = \text{const}$ has the light cone of the fold. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp factor is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$ equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\, g^{ac}\, \partial_\sigma g_{cb} = -H\,\mathrm{sgn}(\sigma)\,\delta^a{}_b \quad \text{(all seven directions of the fold)} \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^a{}_b = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of UM-4; $K = K^c{}_c$ its trace. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.

- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\hat{g}_{ab}$: The metric of the seven-dimensional fold (FD-2) in its coordinates $x_1,\dots,x_7$ (or in its frame), $\hat{g}^{ab}$ its inverse; $\hat{h}_{\mu\nu}$ is its part along the four observable directions. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \dots, x_7$ for a tensor of eight dimensions, $x_1,\dots,x_7$ for a tensor of the fold, $x_1,\dots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.

(a, b frame indices along the fold); it jumps by $-2H$ in every direction of the fold. The field equations of gravity of UM-4, ${\mathcal{E}_{(1)}}^M{}_N + \Lambda\delta^M_N = \kappa{\mathcal{T}}^M{}_N$ (Einstein gravity: the author's Einstein-Lovelock equations with the weights of the Gauss-Bonnet and cubic terms set to zero; $\mathcal{E}_{(1)}$ the Einstein tensor, $\Lambda = -21H^2$ the cosmological constant, $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions, ${\mathcal{T}}^M{}_N$ the energy-momentum tensor, δ^M_N the Kronecker delta), are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_\mathrm{D}(\sigma)$ into them: the metric solves them only if the fold carries a thin source ${\mathcal{S}^a}_b$, an energy-momentum tensor concentrated on $\sigma = 0$, given by the integral of the equations across the fold (the junction condition of Israel),

$$\kappa\,{\mathcal{S}^a}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma\,{\mathcal{E}_{(1)}}^a{}_b = -\Big(\big({K^a}_b\big|_{0^+} - {K^a}_b\big|_{0^-}\big) - {\delta^a}_b\big(K\big|_{0^+} - K\big|_{0^-}\big)\Big) \tag{FD-11}$$

Symbols in (FD-11).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- ${\mathcal{S}^a}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa{\mathcal{S}^a}_b = \int_{0^-}^{0^+}\mathrm{d}\sigma\,{\mathcal{E}_{(1)}}^a{}_b$ (the junction condition of Israel); ${\mathcal{S}^4}_4$ is minus its energy density, ${\mathcal{S}^1}_1$ and ${\mathcal{S}^5}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa{\mathcal{S}^a}_b = -12H{\delta^a}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \dots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1,\dots,7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1,\dots,7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the

quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $K^{a}{}_{b}$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^{a}{}_{b} = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of UM-4; $K = K^{c}{}_{c}$ its trace. Unit: GeV (1/length). Status: DERIVED.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^{M}{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

($K = K^{c}{}_{c}$ the trace of the extrinsic curvature), with $\mathcal{S}^{4}{}_{4} = -$(its energy density), $\mathcal{S}^{1}{}_{1} = \mathcal{S}^{2}{}_{2} = \mathcal{S}^{3}{}_{3}$ its pressure along ordinary space and $\mathcal{S}^{5}{}_{5} = \mathcal{S}^{6}{}_{6} = \mathcal{S}^{7}{}_{7}$ its pressure along the extra times; $\mathcal{S}$ is $\mathcal{T}$ integrated across the fold, an energy per unit volume of the six other directions of the fold (unit GeV^7). For the static block (a_4 and b_4 constant) and, to the first order in the curvature of the fold divided by H^2, for every fold

$$\begin{aligned} &\text{static block:}\quad \kappa\mathcal{S}^{a}{}_{b} = -12H\,\delta^{a}{}_{b}\quad\text{(a pure tension)}, \qquad \kappa\lambda_{\mathrm{F}} = 12H, \\ &\text{any fold:}\quad \kappa\mathcal{S}^{a}{}_{b} = -12H\,\delta^{a}{}_{b} + \frac{2}{5H}\,\hat{G}^{a}{}_{b} + \ldots \end{aligned} \tag{FD-12}$$

Symbols in (FD-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+}\mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^{a}{}_{b}$ (the junction condition of Israel); $\mathcal{S}^{4}{}_{4}$ is minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^{a}{}_{b} = -12H\delta^{a}{}_{b}$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- λ_F: The tension that the static block requires of the fold, $\kappa\lambda_\mathrm{F} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_\mathrm{F} V_*$, and the excess of the total over $\lambda_\mathrm{F} V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_\mathrm{F} V_* = 30 H^2 M_\mathrm{red}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- M_red: The reduced Planck mass, $M_\mathrm{red} = (8\pi G_\mathrm{N})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-4: $\rho_\Lambda = \rho_\mathrm{ff} + \rho_\mathrm{vac} - \lambda_\mathrm{F} V_*$ (four-dimensional normalisation). Unit: GeV^4. Value: $0.685\, \rho_\mathrm{c}^{(0)} = 2.5 \times 10^{-47}$ GeV^4 (Planck 2018). Status: DERIVED.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$ with the Planck mass $M_\mathrm{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2$ GeV/cm^3. Status: STANDARD.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_\mathrm{F} V_* + \rho_\Lambda - \rho_\mathrm{vac}$: together with the vacuum energy ρ_vac of the fields of the fold it makes up the tension $\lambda_\mathrm{F} V_*$ that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_vac: The vacuum energy of the fields of the fold: the minimum of their potential (for the Standard Model the minimum of the Higgs potential with its loop corrections), a tension of the seven-dimensional fold. Unit: GeV^4. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.

($\hat{G}^a{}_b$ the Einstein tensor of the metric FD-2 of the fold): the tension plus matter that obeys the Einstein equations of seven dimensions, $\hat{G}^a{}_b = \kappa_7 \mathtt{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$ ($\mathtt{t}^a{}_b$ the matter of the fold), which for frozen extra times are the Friedmann equations of general relativity. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0 x_0} = \cot^2(6H x_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2 \Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \ldots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one

period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-4 the Higgs doublet (with the author's ad hoc vacuum, a time-dependent choice of its vacuum value), three generations of quarks and leptons, three right-handed neutrinos and the fold fluid (its tension and the pressure along the extra times that keeps their volume fixed) live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \ldots, x_4$ (no field depends on the extra times). The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ live in the bulk; their massless modes are bound to the fold, so that the photon and the gluons of the four-dimensional world are waves of the bulk that the fold holds, and photons that leave the fold travel through the bulk at the speed of light. dirac16complex (the author's spinor field of eight dimensions) and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the states χ of dirac16complex that the fold holds (in UM-4 long-lived resonances) are the part of it near the fold, and the portals (couplings between dirac16complex and the fields of the fold) act at $\sigma = 0$. Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \ldots, x_4$. In UM-4 files "the fold" means this surface; the examples below use the word for other surfaces with the same properties. The word is not used in its other senses: a set of equal energy levels is called degenerate, and a growth by the factor e is called an e-folding.

Examples of folds that are meaningful for UM-4.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the quarks, leptons and the Higgs field, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp factor $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. A compact direction of length L would give every field a tower of particles with the masses $2\pi n/L$, $n = 1, 2, \ldots$ (a Kaluza-Klein tower); because the copies are infinitely far apart, the period $\pi/(6H)$ of x_0 produces no such tower with the spacing $12H$: every field of the bulk has instead a continuum of modes that starts at zero mass, and the fold holds its massless mode.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property 3). In UM-4 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \ldots, x_7$, and the crease is the fold.
4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_\mathrm{D}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- δ_D: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every metric function of FD-4 depends on σ through $|\sigma|$ and is folded at $\sigma = 0$ in this sense. 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_\mathrm{RS}^2 + \mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_\mathrm{RS} = \frac{6k_\mathrm{RS}}{\kappa_5}, \qquad \Lambda_\mathrm{RS} = -\frac{6k_\mathrm{RS}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- y_RS: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-4). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_RS: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp factor of the brane is $\mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}$ (FD-8, an example, not UM-4). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_RS: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_\mathrm{RS} = 6k_\mathrm{RS}/\kappa_5$ (FD-8, an example, not UM-4). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, 8π times the Newton constant of five dimensions (FD-8, an example, not UM-4). Unit: GeV^{-3}. Status: STANDARD.
- Λ_RS: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_\mathrm{RS} = -6k_\mathrm{RS}^2/\kappa_5$ (FD-8, an example, not UM-4). Unit: GeV^5. Status: STANDARD.

has a fold at $y_\mathrm{RS} = 0$ with the properties 1 to 3 (its coordinate y_RS is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_RS (Israel's junction condition) and in the bulk the vacuum energy Λ_RS, and the bulk is otherwise empty. The fold of UM-4 is of the same kind, with every direction of the fold warped by the same factor as in the model of Randall and Sundrum: its static block is a space of constant curvature $-H^2$ in all eight directions, a vacuum of Einstein gravity with $\Lambda = -21H^2$, with a fold of pure tension (FD-12), and the graviton is bound to the fold as there. It differs in that the fold has seven dimensions, three of them extra times, in that the space is not cut at the brane (Randall and Sundrum identify the two sides of their brane, one physical side; the fold of UM-4 has two distinct sides, example 7), and in that the gauge fields live in the bulk as well. 7. *A particle held by a thin layer.* A particle held by an attractive thin layer has a wave function that falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-4 the states χ of dirac16complex near the fold are of this kind only near the fold: at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{(7/2)H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega x_4}$)

$$\chi(\sigma)=\mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+)\ \ (0<\sigma\ll\sigma_\mathrm{t}),\qquad \chi(\sigma)=\mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-)\ \ (0<-\sigma\ll\sigma_\mathrm{t}),$$
$$\chi(0^+)=\exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L}P_\mathrm{L}+\mu_\mathrm{R}P_\mathrm{R}\big)\Big)\chi(0^-),\qquad \omega_0=\omega_\mathrm{r}-\frac{\mathrm{i}}{2}\,\Gamma_\chi,\qquad \omega_\mathrm{r}\to\omega_\mathrm{b}=\sqrt{m_\Psi^2-q_\mathrm{b}^2} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi=\mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex at the fold: inside the turning point its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- σ_t: The turning point of a state of dirac16complex of the frequency ω: the depth at which the warped frequency equals the bulk mass, $|\omega|\mathrm{e}^{H\sigma_\mathrm{t}}=m_\Psi$; beyond it every solution oscillates. Unit: length. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M=\mathbf{e}_a{}^M T^a$, $\gamma_M=g_{MN}\gamma^N$; $\gamma^{(a)}=T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L}+\mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L}=(\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R}=(\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- ω_0: The energy at rest (zero momentum along ordinary space and along the extra times) of a state of dirac16complex at the fold; complex for a resonance, $\omega_0=\omega_\mathrm{r}-\frac{\mathrm{i}}{2}\Gamma_\chi$. Unit: GeV. Status: DERIVED.
- ω_r: The mass of the particle χ: the real part of the energy at rest of the resonance of dirac16complex at the fold, $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))+\ldots$. Unit: GeV. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- Γ_χ: The width of the particle χ at rest: the rate of its decay into the waves of dirac16complex in the bulk. Unit: GeV. Value: below $1/t_0=1.5\times10^{-42}$ GeV for dark matter. Status: DERIVED.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H\to0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))+\ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8=T_0T_1\cdots T_7$ (the author's T16^A[8]) $=\mathrm{diag}(-\mathbf{1}_8,\mathbf{1}_8)$; $\Gamma_8^2=\mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi}=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN}=\eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN}=\eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto\mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x}=(x_1,x_2,x_3)$. Unit: length. Status: AUTHOR.
- t_0: The age of the universe. Unit: time. Value: 13.8 Gyr. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16}=T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.

- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.

where μ_{L}, μ_{R} are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the fields of the fold), $q_{\mathrm{b}} = -m_\Psi(\mu_{\mathrm{L}} + \mu_{\mathrm{R}}) \tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_{\mathrm{L}}\mu_{\mathrm{R}}}$ ($q_{\mathrm{b}} = -m_\Psi\mu_{\mathrm{L}}/2$ for $\mu_{\mathrm{R}} = 0$), and ω_{r} tends to the energy ω_{b} of the bound state of UM-3 when $H/q_{\mathrm{b}} \to 0$. Because the time is warped, the energy ω of the field enters its equation multiplied by $\mathrm{e}^{H|\sigma|}$: beyond the turning point σ_{t} ($\omega_{\mathrm{r}}\mathrm{e}^{H\sigma_{\mathrm{t}}} = m_\Psi$) the profile oscillates as a wave that leaves the fold, so the state is a resonance of the width Γ_χ, which falls like $\mathrm{e}^{-2F(\upsilon_{\mathrm{b}})m_\Psi/H}$ with $F(\upsilon) = \ln((1+\sqrt{1-\upsilon^2})/\upsilon) - \sqrt{1-\upsilon^2}$ and $\upsilon_{\mathrm{b}} = \omega_{\mathrm{b}}/m_\Psi$ (the tunnelling through the region between the fold and σ_{t}). The profile is not a function of $|\sigma|$: it jumps at the fold, $\chi(0^+) \neq \chi(0^-)$. 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the states at the fold of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma)$ (FD-10): it maps the states at the fold of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of a mean field of a gas of the particles χ that is the same on both sides of the fold: the four resonances of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so such a mean field cannot mix the two signs of the indefinite charge. R_2 maps the states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory (the eleven-dimensional theory whose limits include the superstring theories) on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); each carries one E_8 gauge group while gravity fills the bulk. As in UM-4, gravity fills the bulk; unlike UM-4, the gauge fields live on the mirror planes (in UM-4 they live in the bulk), and the two sides of each plane are identified (one physical side).

What is not a fold.

- *The tips* $z = 0$ and $z = \pi$ (mod π): the mirror exchanges them within a period, but there the warp factor vanishes, $g_{x_0x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp factor*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma = 0$, but without a kink and without a thin source.

2 How to read this document

This document records the origin of everything in the universe model UM-4: the author's inputs and orders, the decisions taken while building the model, the computations that verify every derived statement (each a Python script that checks itself), the external measurements and software that were used, and the commands that reproduce

every result and every document. It is a record of origins, decisions and computations, not of the model itself, and so it contains no equations other than those of the definition of the fold at its beginning.

The status labels used throughout UM-4: **AUTHOR** (from the author), **RECORD** (from the record of this repository), **STANDARD** (textbook physics or a published measurement), **ASSUMED** (a choice without derivation), **INVENTED** (a new interaction: the portals of dirac16complex), **DERIVED** (computed and verified), **ESTIMATE** (an order of magnitude), **OPEN** (not settled).

UM-4 is the successor of the universe model UM-3. The author's order of 2026-10-25, 13:15 PDT, which made UM-4, says "keep UM-3 and create a new UM-4, placed in new folders": UM-3 is kept unchanged in the folder `universe_model_3` of the repository, as committed in `c5a93c4`, and everything of UM-4 is new, in the folder `universe_model_4`, with programs of its own (`um4_*.py`) and documents of its own.

3 The author's inputs

input	content and place
the order of 2026-10-24, 17:02 PDT	stop every earlier stage permanently; a new revision folder; read `4+4.nb` and verify its eight real 16×16 Dirac matrices; the metric with b_4^2 (b_4 a function of the time x_4 in the extra-time components of the metric, never identically 1); the complete coupled field equations of the listed fields (Einstein-Lovelock gravity, the Higgs doublet with the ad hoc vacuum, three generations of lepton doublets, the right-handed electron, muon and tau, W^1, W^2, W^3, B, three generations of quark doublets, the gluons, dirac16complex coupled to gravity with a non-trivial spin connection and with its left- and right-handed halves coupled to every Standard-Model field, only the left-handed half to the Higgs field); new phenomenological interactions; an accurate approximate Kohn-Sham solution; the CMB, the CNB and the BAO; the x_0 mass mechanism; dark matter and dark energy; every equation and every symbol in every document, never a reference to another file; Markdown, LaTeX and PDF; push as soon as verified. In the author's private prompt file (not committed).
the notebook 4+4.nb	the author's notebook "eight, real-valued, 16 by 16, Dirac Matrices" (Wolfram 15.0) with the matrices `T16^A[0..8]`; kept unchanged in the folder `universe_model_3/author_inputs` (its sha256 is given below) and read from there by the programs of UM-4 (not copied)
the notebook of the record	`Dirac_Einstein-Lovelock_4+4_Nash.nb` in the root of the repository: the coordinates $X = \{x_0, \ldots, x_7\}$ (Section DEFCHART: x_0 the hidden direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the extra times) and the Einstein-Lovelock action with the weights $(1, 1/32, 1/1152)$ of the Einstein, Gauss-Bonnet and cubic terms and the cosmological-constant parameter $\Lambda_{\mathrm{N}} = 10$ (In[134]); UM-4 uses this action with the weights of the Gauss-Bonnet and cubic terms set to zero, Einstein gravity with $\Lambda = -21H^2$
the decision of 2026-10-24, 17:32	"rather have three ν_{R}": three right-handed neutrinos, Dirac masses (in the conversation)
the reason of 2026-10-24, 17:34	Gogilashvili and Tamborra, "Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504: why the flavour conversion must be correct (a third-party paper on the author's computer, read, not committed)
the standing rule of 2026-10-14	no single calculation may take longer than 423 s (7 min 3 s); every command runs under a time limit (recorded in `claude_failures.md` and `HANDOFF.md` of the repository)

input	content and place
the orders of 2026-10-25, 03:21, 03:23, 03:54, 05:09 and 07:55 PDT	given for UM-3 and carried over to UM-4: the sources on the fold (a thin fluid and the gravity of the fold); the answers to the open questions (the Krein norm, fields that depend on the extra times, particles faster than light, the primordial era, the lightest neutrino mass and the Dirac phase); one definition of the fold, with its examples, in every file that uses the word; proper attribution of the stabiliser of the bulk of UM-3 to Claude (UM-4 has no stabiliser); "Gravitons must travel at c ... my choices are wrong, and you MUST correct them"
the order of 2026-10-25, 13:15 PDT	"Since in the UM-3 model anything that propagates in the bulk outruns light on the fold, and bulk scalars and photons faster than light, my model is physically incorrect. You must find a new metric model in which x4 is warped like the other directions, both photons and gravitons actually propagate at c in the bulk, photon waves are transverse spin-1 phenomena, Gravitational waves (ripples in spacetime) are spin-2 phenomena (because they originate from the rank-2 stress-energy tensor) and possess the following Polarization Modes: the spin-2 nature of gravitational waves manifests physically as characteristic"plus" (+) and "cross" (×) polarization modes that stretch and squeeze space perpendicularly. Find the best model and re-calculate and re-document; however keep UM-3 and create a new UM-4, placed in new folders."

The sha256 of the author's notebook `4+4.nb` as received and committed:

`7ec1709a0028bce0b443f116cf27b5d3e670de7aa195659a9c3274527270429d`

The author's private files (his prompt files, his notes, the third-party paper, his scratch notebook of 2026-10-16 with its `.mx` files and the files `Generalized_Kronecker_Delta.*`) are never committed; the repository's `.gitignore` excludes them.

4 The decisions taken in UM-4

decision	reason	status
the time x_4 warped by the factor $(\sin^2 z)^{1/6}$ of the other six directions of the fold, everything else of the author's metric kept	the order of 13:15; the smallest change of the author's metric that meets it: with the same factor the light cone of every slice of the bulk is that of the fold, and among the static warps of the author's form (ordinary space, the time and the extra times warped at their own rates) only equal rates give a vacuum of gravity with a cosmological constant (`um4_geometry.py`, G8 and G10)	AUTHOR (the order), ASSUMED (the way it is met), DERIVED (its properties)
Einstein gravity with $\Lambda = -21H^2$	the only choice of the author's Lovelock weights for which gravitons travel on the light cone on every background, not only around the static block (`um4_gravity.py`, L1 to L4); with it the static block is a space of constant curvature $-H^2$, a vacuum without a stabiliser, and the fold a pure tension	DERIVED

decision	reason	status
the gauge fields of the Standard Model in the bulk	the order puts the photon in the bulk, and with it the electroweak fields of which it is a combination; the gluons are put there as well	AUTHOR (the photon), ASSUMED (all gauge fields)
the Higgs doublet, the quarks and the leptons on the fold, dirac16complex in the bulk	the author's specification of dirac16complex as a field of eight dimensions; the fields of the Standard Model other than the gauge fields are not described in the bulk	ASSUMED
right-handed quark singlets and Yukawa couplings	masses of the quarks and cancellation of the anomalies (defaults of the notification of 2026-10-24, 17:20; the author answered only on the neutrinos)	ASSUMED
three right-handed neutrinos	the author's decision of 2026-10-24, 17:32	AUTHOR
the fold fluid	the kink of the warp at the fold needs a thin source: a pure tension $\kappa\lambda_{\mathrm{F}} = 12H$ at the static block, and a pressure q_{hold} along the extra times that holds their volume fixed (a free volume would change Newton's constant 8500 times faster than lunar laser ranging allows); its value today depends on the pressure of the baryons along the extra times, taken as zero	DERIVED (its values), ASSUMED (the medium, its holding pressure and the pressure of the baryons), OPEN (the origin of the holding pressure, the tuning of the tension)
the bound $H > 2534$ GeV used in every document	the invisible width of the Z boson, whose decay into the bulk the gauge fields of the bulk allow (`um4_propagators.py`, P5); the other bounds (contact interactions, the parameter rho, Newton's law) are weaker	ESTIMATE
the portals of dirac16complex	the author's request for couplings of both halves to every Standard-Model field, only the left-handed half to the Higgs field	INVENTED
the pure-gauge W^2 of the ad hoc vacuum	the Gauss law of $SU(2)$; in UM-4 it is the value at the fold of the field of the bulk	DERIVED
no dependence of any field on the extra times	such fields are tachyonic; the restriction is consistent but unstable, and a gauge principle that removes the momentum along the extra times is needed	ASSUMED (OPEN: the gauge principle)
the parameters of the Kohn-Sham computation ($m_\Psi/H = 10^4$, $\mu_{\mathrm{L}} = -0.6$, $\mu_{\mathrm{R}} = 0$, the box $\lvert\sigma\rvert \le 40/m_\Psi$ with the step $0.03/m_\Psi$; the weakly bound point $q_{\mathrm{b}} = 0.03\,m_\Psi$, $H = 10^{-7}m_\Psi$)	the states of dirac16complex at the fold are resonances, so the gas is computed deep in the regime of long-lived particles, far inside the turning point	ASSUMED

decision	reason	status
the lightest neutrino mass zero, the Dirac phase zero	not fixed by the specification; the phase drops out of every result of UM-4, and DESI DR2 with the CMB and NuFIT 6.0 bounds the lightest mass below 23 meV	ASSUMED (the mass within the bound), DERIVED (the phase is irrelevant)
the primordial spectrum, the optical depth and the densities from Planck 2018	the author's vacuum is not a solution; the primordial era of UM-4 can be an inflation of the fold with frozen extra times, which needs a Weyl tensor of the bulk and a rate below $H/10$; UM-4 does not specify its inflaton	ASSUMED (inputs; the inflaton OPEN)
the abundance of the dark matter taken from the measurement	the particles χ of UM-4 are too heavy to be a thermal relic	ASSUMED (the origin OPEN)
the quantisation of dirac16complex with positive energies	it makes the vacuum stable; the modes of positive Krein charge are then ghosts, which the chiral couplings of the fold produce; the classical stability of the gas is not decided	DERIVED (the consequences), OPEN (the stability of the gas, a unitary coupling)
one definition of the fold in every file	the order of 2026-10-25, 03:54: the definition with its examples, `docs_src/fold_definition.md` for the documents and `docs_src/fold_definition.txt` for the code and the JSON files, inserted and checked by `tools/fold_block.py`; the repository files `README.md`, `HANDOFF.md` and `claude_failures.md` carry the definition of UM-4 as plain text next to that of UM-3	AUTHOR (the order)
the Kohn-Sham module in parts	the standing rule of 423 s: the module runs in four parts with checkpoints, each under the time limit, and an assembly that makes the checks	AUTHOR (the rule)
the lunar-laser-ranging bound of L. Biskupek, J. Müller and J.-M. Torre (2021)	the bound on the change of Newton's constant used in every document of UM-4 (two standard deviations)	STANDARD

5 The computations

Every derived statement rests on one of these scripts in `universe_model_4/code/`. Each computes, checks itself and writes its results to `universe_model_4/outputs/` and its checks to `universe_model_4/reports/`.

script	what it computes	checks
`um4_common.py`	the report of the checks (a library)	
`um4_gammas.py`	the author's Dirac matrices, rebuilt from `4+4.nb` (a library)	

script	what it computes	checks
`um4_lovelock.py`	the Lovelock tensors from the frame curvature (a library)	
`um4_dirac_matrices.py`	reality, Clifford relations, Spin(4,4) and Pin(4,4), chiral projectors, agreement with every stored output of the notebook and with the earlier records of the repository	55
`um4_dirac_matrices_doc.py`	the document `dirac matrices` from the checked results	
`um4_geometry.py`	the proper chart with the warped time, the exact curvature of the metric of UM-4, the static block of constant curvature, its Lovelock tensors, the exact bulk equations of Einstein gravity, the Ricci-flat folds, the pure tension, the light cone of every slice, the spin connection with the author's matrices, the static warps and their vacua	10
`um4_gravity.py`	the vacua of the static family for any weights, the graviton around the static block for any weights, the light cone of the gravitons on every warped background, the choice of Einstein gravity	4
`um4_bulk.py`	the modes of the scalar and of the gauge field of the bulk, the photon as a transverse wave of spin 1, the graviton of the bulk and the gravitational wave with its plus and cross polarisations, the bulk fermion, the boost invariance of dirac16complex, the speeds, the speeds of UM-3 recomputed, the continuum of massive modes from zero mass and its spectral density at the fold	10
`um4_fold.py`	the bulk metric of an expanding fold to the first order, the fold as a seven-dimensional Einstein brane, the exact equations of the fold, the Friedmann equations with frozen extra times, the size of the corrections, the Weyl term that dust requires	6
`um4_foldsources.py`	the tension, the gravitational wave of the fold, the induced gravity, the tuning of the tension, the fold fluid	5
`um4_background.py`	the vacua of the author's form, the expanding fold, the free volume excluded by lunar laser ranging, the fold factor, the ad hoc vacuum as a gauge rotation and its numbers	6

script	what it computes	checks
`um4_propagators.py`	the propagators of the bulk from one point of the fold to another, their expansions, Newton's and Coulomb's laws, the W and Z resonances, the bounds on H	5
`um4_causality.py`	no shortcut through the static bulk, the light cone of an expanding fold, the largest advance, the closed time-like curves	4
`um4_fields.py`	the bilinears of dirac16complex, the Krein operator, the electroweak vacuum with the ad hoc component, the anomalies	16
`um4_spectra.py`	the resonances of dirac16complex at the fold with the exact matrices, their widths, their dispersion, the lifetime condition, their Krein signature and their mirror (reads the outputs of `um4_propagators.py`)	8
`um4_kohnsham.py`	the self-consistent Kohn-Sham states of a gas of the particles χ, a weakly bound point, how dilute the gas of the universe is	10
`um4_krein.py`	the Krein form and its quantisation, the symmetries and mirrors of the states at the fold, the flat directions of the gas and its truncated response, the partner level, the selection rules at the fold, stability and unitarity (reads the snapshots of `um4_kohnsham.py`)	7
`um4_extratimes.py`	fields with a momentum along the extra times: the dispersion, the states of dirac16complex with such a momentum, the growth with frozen extra times, compact extra times and their harmonics, the truncation	5
`um4_neutrino.py`	the flavour blindness of the portals, the matter conversion, the crossing criterion, the equipartition, the cosmic neutrino background (CNB)	5
`um4_flavour.py`	the neutrino flavours in the Sun and in supernovae	4
`um4_numass.py`	the lightest neutrino mass and the Dirac phase: the masses of both orderings, the bounds of DESI DR2 and KATRIN, the phase in the results of UM-4, the effect of the lightest mass, the record against NuFIT 6.0	5

script	what it computes	checks
`um4_thermal.py`	the degrees of freedom, the relic density, the portals and direct detection, the right-handed neutrinos, the ad hoc vacuum before nucleosynthesis, no thermal relic, the binding by the Higgs portal and the lifetime (reads the outputs of `um4_spectra.py`)	8
`um4_primordial.py`	the author's vacuum (not a solution), the exact de Sitter fold, the inflation with frozen extra times, its scales, Higgs inflation (an extension of the action), the fold factor	6
`um4_cosmology.py`	the cosmic microwave background (CMB), the CNB and the baryon acoustic oscillations (BAO) with the Boltzmann code CAMB and an independent background, against Planck 2018 and DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument)	6

The 20 programs have 185 checks, all passed. The documents are written in Markdown and built with `universe_model_4/tools/build_docs.py` (pandoc to LaTeX, pdflatex to PDF, deterministic). `universe_model_4/docs_src/registry.py` holds the meaning, unit, value and status of every symbol; `universe_model_4/tools/make_inventory.py` writes the inventory of the tagged equations from `FIELD_EQUATIONS.md`, and `universe_model_4/tools/symbols_tool.py` copies every tagged equation into the other documents (`sync`), writes the symbol list after every display and the list of further symbols at the end of every document (`expand`) and checks that every symbol of every display is defined, that every symbol used in the meaning of a listed symbol is listed in the same document, that every tagged equation that a document cites is displayed in it, and that every tagged equation is written identically in every document (`check`). `universe_model_4/tools/symbols_close.py` adds to the lists of a document the entries that the meanings of its listed entries need (with `universe_model_4/tools/closure_check.py`), and `universe_model_4/tools/inline_symbols.py` checks that the symbols of the inline formulas of the prose are listed in the document as well (adding the missing entries to the list of further symbols; a symbol without an entry, such as a label value or a generic letter of an example, is defined in words where it is used). `universe_model_4/tools/fold_block.py` inserts the one definition of the fold into every file that uses the word (`insert`) and checks that every such file carries it word for word (`check`). `universe_model_4/tools/nb_reader.py` reads the cells of the author's notebooks without Mathematica.

6 External measurements and software

source	used for
Planck Collaboration, Planck 2018 results VI, A&A 641, A6 (2020), Table 2 (TT,TE,EE+lowE+lensing)	the cosmological parameters and the derived quantities of the CMB
Planck 2018 plik-lite likelihood files (Planck Legacy Archive), via github.com/heatherprince/planck-lite-py (Prince and Dunkley 2019)	the CMB spectra (downloaded at run time into `universe_model_4/external/`, not redistributed)
DESI Collaboration, DESI DR2 Results II, arXiv:2503.14738, Table IV	the BAO distances; the 2.3 and 3.1 standard-deviation statements of its abstract

source	used for
W. Elbers et al. (DESI Collaboration), arXiv:2503.14744, section IV C	DESI DR2 with the CMB combined with NuFIT 6.0 (95 per cent, cosmological-constant model): the lightest mass below 0.023 eV (normal ordering) and 0.024 eV (inverted), the sum below 0.101 eV and 0.133 eV, the probability 0.91 of the normal ordering; the bound 0.0642 eV on the sum of three equal masses
ACT Collaboration (Atacama Cosmology Telescope), DR6 constraints on extended cosmological models, arXiv:2503.14454; T. Louis et al. (ACT Collaboration), arXiv:2503.14452	the effective number of neutrino species $N_{\rm eff} = 2.89 \pm 0.11$, the sum of the neutrino masses below 0.082 eV; the spectral index 0.974 ± 0.003 (ACT DR6 with Planck, CMB lensing and DESI)
Planck Collaboration, Planck 2018 results X, A&A 641, A10 (2020); M. Tristram et al., Phys. Rev. D 105, 083524 (2022) (Planck PR4 with BICEP/Keck 2018 and BAO)	the spectral index 0.9649 ± 0.0042; the bound $r_{\rm T} < 0.032$ (95 per cent) on the tensor-to-scalar ratio
P. Janot and S. Jadach, Phys. Lett. B 803, 135319 (2020)	the number of light neutrinos 2.9963 ± 0.0074 measured at the Z boson, which bounds the decay of the Z boson into the bulk ($H > 2534$ GeV)
ATLAS Collaboration, JHEP 11 (2020) 005	the limit $\Lambda_{\rm CI} > 26.0$ TeV on the scale of contact interactions of electrons and muons ($H > 1.3$ TeV)
Particle Data Group, Review of Particle Physics (2022 and 2024)	the masses of the Higgs boson (125.25 GeV), of the quarks and charged leptons and of the nucleon, the Fermi constant 1.1663788×10^{-5} GeV^{-2} and $v = 246.22$ GeV, the fine-structure constant; the parameter rho of the electroweak fits, 1.00031 ± 0.00019 (2024)
J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer and B. R. Heckel, Phys. Rev. Lett. 124, 101101 (2020)	Newton's law tested by torsion balances down to 52 micrometres
B. P. Abbott et al. (LIGO Scientific, Virgo, Fermi GBM and INTEGRAL Collaborations), Astrophys. J. Lett. 848, L13 (2017)	the speed of gravitational waves, $-3 \times 10^{-15} < c_{\rm g} - 1 < 7 \times 10^{-16}$ (GW170817 and GRB 170817A)
L. Biskupek, J. Müller and J.-M. Torre, Universe 7, 34 (2021)	the bound $(-5.0 \pm 9.6) \times 10^{-15}$ per year of lunar laser ranging on the change of Newton's constant, against a free volume of the extra times
W. Israel, Nuovo Cimento B 44, 1 (1966); T. Shiromizu, K. Maeda and M. Sasaki, Phys. Rev. D 62, 024012 (2000); L. Randall and R. Sundrum, Phys. Rev. Lett. 83, 4690 (1999)	the junction condition of the fold; the exact equations of the fold in a bulk of constant curvature; the brane in anti-de Sitter space whose tension and fine tuning the fold of UM-4 shares
A. D. Sakharov, Dokl. Akad. Nauk SSSR 177, 70 (1967); D. V. Vassilevich, Phys. Rep. 388, 279 (2003); S. N. Solodukhin, arXiv:1502.03758	induced gravity; the heat-kernel coefficients of scalars, fermions and gauge fields
K. Griest and M. Kamionkowski, Phys. Rev. Lett. 64, 615 (1990)	the unitarity bound on the mass of a thermal relic
LUX-ZEPLIN (LZ) Collaboration, Phys. Rev. Lett. 135, 011802 (2025), arXiv:2410.17036 version 3 (4.2 tonne-years)	the limit 2.2×10^{-48} cm^2 at 40 GeV (90 per cent confidence) on spin-independent scattering
Borexino Collaboration, Nature 562, 505 (2018)	the survival probabilities of solar neutrinos
the global fit NuFIT 5.2 of the neutrino oscillation data (Esteban, Gonzalez-Garcia, Maltoni, Schwetz and Zhou, JHEP 09 (2020) 178; version 5.2 of November 2022 at www.nu-fit.org), normal ordering, the fit without the Super-Kamiokande atmospheric data	$\sin^2\theta_{12} = 0.303$, $\sin^2\theta_{13} = 0.02203$, $\sin^2\theta_{23} = 0.572$, $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.511 \times 10^{-3}$ eV^2

source	used for
the global fit NuFIT 6.0 (I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro and T. Schwetz, JHEP 12 (2024) 216, arXiv:2410.05380), data set IC24 with the Super-Kamiokande atmospheric data, and IC19 without it ($\sin^2\theta_{12} = 0.307$, $\sin^2\theta_{23} = 0.561$, $\sin^2\theta_{13} = 0.02195$, $\delta_{\mathrm{CP}} = 177°$, $\Delta m^2_{31} = 2.534 \times 10^{-3}$ eV^2)	$\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\mathrm{CP}} = 212°$, $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV^2, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV^2 (normal ordering), and the inverted ordering, for the comparison with the record
KATRIN Collaboration, Science 388, 180 (2025)	the limit 0.45 eV (90 per cent) on the effective electron-neutrino mass of tritium beta decay
Long, Lunardini and Sabancilar, arXiv:1405.7654	the capture rate of the CNB on tritium (4 and 8 per year per 100 g for Dirac and Majorana neutrinos)
Gogilashvili and Tamborra, arXiv:2605.16504 (version 3)	the flavour equipartition (the equal sharing of the three flavours that fast flavour conversion produces) and its effect on the explosions of massive stars
Bahcall's exponential solar density profile; Petcov's crossing probability; Izaguirre, Raffelt and Tamborra's dispersion relation; Morinaga's crossing theorem; Ma and Bertschinger's Boltzmann equations; F. Bezrukov and M. Shaposhnikov, Phys. Lett. B 659, 703 (2008) (Higgs inflation)	the standard physics named in the documents
CAMB 2.0.5 (Lewis, Challinor and Lasenby 2000)	the Boltzmann hierarchy and the CMB spectra
Python 3.13.17 with numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0, matplotlib 3.11.2; pandoc 3.1.3; pdfTeX (TeX Live 2023)	every computation and document

7 How to reproduce

From the root of a clone of the repository, with the software above installed, each command run under `timeout 423`:

```
python universe_model_4/code/um4_dirac_matrices.py
python universe_model_4/code/um4_dirac_matrices_doc.py
python universe_model_4/code/um4_geometry.py
python universe_model_4/code/um4_gravity.py
python universe_model_4/code/um4_bulk.py
python universe_model_4/code/um4_fold.py
python universe_model_4/code/um4_foldsources.py
python universe_model_4/code/um4_background.py
python universe_model_4/code/um4_propagators.py
python universe_model_4/code/um4_causality.py
python universe_model_4/code/um4_fields.py
python universe_model_4/code/um4_spectra.py
python universe_model_4/code/um4_kohnsham.py part K1
python universe_model_4/code/um4_kohnsham.py part I
python universe_model_4/code/um4_kohnsham.py part K6
python universe_model_4/code/um4_kohnsham.py part K9
python universe_model_4/code/um4_kohnsham.py assemble
python universe_model_4/code/um4_krein.py
python universe_model_4/code/um4_extratimes.py
python universe_model_4/code/um4_neutrino.py
python universe_model_4/code/um4_flavour.py
python universe_model_4/code/um4_numass.py
python universe_model_4/code/um4_thermal.py
python universe_model_4/code/um4_primordial.py
```

```
python universe_model_4/code/um4_cosmology.py
python universe_model_4/docs_src/registry.py
python universe_model_4/tools/fold_block.py insert
python universe_model_4/tools/make_inventory.py
python universe_model_4/tools/symbols_tool.py sync
python universe_model_4/tools/symbols_tool.py check
python universe_model_4/tools/fold_block.py check
python universe_model_4/tools/build_docs.py
```

Every script prints one line per check and ends with "n/n checks passed". `um4_spectra.py` reads the outputs of `um4_propagators.py`, `um4_thermal.py` those of `um4_spectra.py`, `um4_krein.py` the snapshots of the parts of `um4_kohnsham.py`, `um4_dirac_matrices_doc.py` the outputs of `um4_dirac_matrices.py`, and `um4_cosmology.py` downloads the Planck files once (about 7 MB). By the standing rule no single calculation may take longer than 423 seconds: a part of `um4_kohnsham.py` that reaches its own limit of 400 seconds saves its state and exits with the code 3, and is then simply run again until it exits with 0. The documents are rebuilt identically (no dates in the LaTeX, fixed PDF metadata).

8 Further symbols

Further symbols. These symbols appear in the definitions of the symbols listed after the equations of this document.

- M_{red}: The reduced Planck mass, $M_{\mathrm{red}} = (8\pi G_{\mathrm{N}})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- ν_α, ν_i: A neutrino field: ν_α the flavour states (the flavour index runs over the electron, muon and tau flavours), ν_i the mass states ($i = 1, 2, 3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i} W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m^2_{ij} = m^2_i - m^2_j$ the squared-mass difference of the states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m^2_2 - m^2_1$, multiplied by v^2_{loc}/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m^2_0 - \mathrm{i} m_0 \Gamma_{\mathrm{bulk}}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\, G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m^2_h/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(k x_3 - \omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- j: A second generation or mass-state index, or a second direction of ordinary space (as i). Status: STANDARD.
- G_{F}: The Fermi constant, $G_{\mathrm{F}} = 1/(\sqrt{2}\, v^2)$ in the Standard-Model vacuum (its measured value); in the equations of the neutrino flavours it stands for the local Fermi constant of the ad hoc vacuum. Unit: GeV^{-2}. Value: 1.1664×10^{-5} GeV^{-2}. Status: STANDARD.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- Γ_{bulk}: The width of the W or the Z boson into the bulk (its decay into the massive modes of the gauge fields of the bulk), $\Gamma_{\mathrm{bulk}} = m_0 (m_0/H)^3/3$. Unit: GeV. Value: 1.4 MeV for the Z and 0.9 MeV for the W at $H = 2.5$

TeV. Status: DERIVED.

- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- g_1: The $U(1)_Y$ gauge coupling of the four-dimensional world (usually g'; not normalised for grand unification): the coupling of the massless mode of the hypercharge field of the bulk, $g_1^2 = 3H\mathsf{g}_1^2/(2V_*)$. Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathsf{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- m_0: The mass of the W or the Z boson without the massive modes of the bulk (the mass of the massless mode with the Higgs vacuum), $m_0^2 = \frac{3}{2}HM_{\rm loc}^2$; $m_0^2 = g_2^2(v^2 + \mathcal{A}^2)/4$ for the W. Unit: GeV. Status: DERIVED.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m_{21}^2 = 7.41\times10^{-5}$ eV^2, $|\Delta m_{31}^2| = 2.511 \times 10^{-3}$ eV^2 in the computations of UM-4 (the global fit NuFIT 5.2 of the oscillation data, November 2022), 7.49×10^{-5} and 2.513×10^{-3} eV^2 in NuFIT 6.0 (2024); the lightest mass $m_1 = 0$ in the computations, below 23 meV by DESI DR2 with the CMB and NuFIT 6.0. Status: STANDARD.
- m_h: The mass of the Higgs boson. Unit: GeV. Value: 125.25 GeV (Particle Data Group 2022). Status: STANDARD.
- $v_{\rm loc}$: The local value of the electroweak vacuum, $v_{\rm loc}^2 = 2\langle\Phi^\dagger\Phi\rangle$: it sets the local masses of the quarks and leptons (each mass times $v_{\rm loc}/v$) and the local Fermi constant; $v_{\rm loc} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- $M_{\rm loc}$: The mass term that the Higgs vacuum on the fold gives a gauge field of the bulk: $M_{\rm loc}^2\delta_{\rm D}(\sigma)$ in its profile equation; for the W, $M_{\rm loc}^2 = \mathsf{g}_2^2(v^2 + \mathcal{A}^2)/(4V_*)$. Unit: GeV$^{1/2}$ ($M_{\rm loc}^2$ in GeV). Status: DERIVED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- g_1, g_2, g_3: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathsf{g}_k^2/(2V_*)$. Unit: GeV^{-2} ($\mathsf{g}^2$: length^4). Status: ASSUMED.
- h_{ij}: The transverse traceless perturbation of the metric of ordinary space by a gravitational wave as the four-dimensional world sees it: $\hat{h}_{ij} = \mathrm{e}^{2a_4}(\delta_{ij} + \mathsf{h}_{ij})$ ($i, j = 1, 2, 3$). Unit: dimensionless. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc, $\mathrm{e}^{-2a_{40}x_4}$ (written Exp[-2 a40 x4] in his notebook), has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-4 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- $N_{\rm eff}$: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- $\mathcal{T}$: The energy-momentum tensor of any one term of the fold (in $\mathcal{T}^r{}_r = \mathcal{L}_{\rm fold}$). Unit: GeV^4. Status: STANDARD.
- $\mathcal{W}$: The warp factor: in a warped metric, the function of the position along one direction that multiplies the squared steps of other directions (its square root multiplies their lengths); in UM-4 the function of σ that multiplies all seven directions of the fold, ordinary space, the time and the extra times, $\mathcal{W} = \mathrm{e}^{-2H|\sigma|} = (\sin^2 z)^{1/6}$, equal to 1 on the fold. Unit: dimensionless. Status: DERIVED.
- $c_{\rm g}$: The speed of gravitational waves on the fold, in units of the speed of light. Unit: dimensionless. Value: 1 at every wavelength for the four-dimensional graviton on the static block and on every fold with frozen extra times, and for the massless graviton of the bulk at every depth; measured $-3 \times 10^{-15} \le c_{\rm g} - 1 \le 7 \times 10^{-16}$ (GW170817). Status: DERIVED.
- $\delta_{\rm CP}$: The Dirac phase of the mixing matrix $U_{\rm PMNS}$. Unit: degree. Value: 212° (NuFIT 6.0, normal ordering); 0 in the computations of UM-4. Status: STANDARD.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein

equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.

- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- q_hold: The pressure along the extra times that the fold fluid adds to its tension to hold the volume of the extra times fixed (four-dimensional normalisation). Unit: GeV^4. Value: $-(\rho - 3\mathcal{P})/2 - q_\mathrm{SM} - q_\chi - q_\mathrm{P}$: 0 for radiation, $-\rho/2$ for dust, $-\rho$ for a vacuum energy; today $-0.842\,\rho_\mathrm{c}^{(0)}$ if the baryons have no pressure along the extra times (ASSUMED). Status: DERIVED (the value), ASSUMED (that the fold fluid supplies it).
- r_T: The ratio of the amplitudes of the primordial spectra of gravitational waves (tensor) and of curvature (scalar) at the pivot scale. Unit: dimensionless. Value: below 0.032 (Planck PR4 with BICEP/Keck 2018 and BAO, M. Tristram et al. 2022, 95 per cent). Status: STANDARD.
- $\mathsf{t}^a{}_b$, t: The matter of the fold above its tension: the thin source minus the tension, $\mathcal{S}^a{}_b = -\lambda_\mathrm{F}\delta^a{}_b + \mathsf{t}^a{}_b$ (eight-dimensional normalisation); $\mathsf{t} = \mathsf{t}^c{}_c$ its trace; with frozen extra times $\mathsf{t}^a{}_b = \mathrm{diag}(P_7, P_7, P_7, -\rho_7, q_7, q_7, q_7)$. Unit: GeV^7. Status: DERIVED.
- θ_{12}, θ_{13}, θ_{23}: The three mixing angles of U_PMNS (its fourth parameter is the Dirac phase, which does not enter the electron-flavour results of UM-4). Unit: dimensionless. Value: 33.40°, 8.54°, 49.14° in the computations of UM-4 ($\sin^2\theta_{12} = 0.303$, $\sin^2\theta_{13} = 0.02203$, $\sin^2\theta_{23} = 0.572$: NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); 33.71°, 8.56°, 43.28° in NuFIT 6.0 (2024, with the Super-Kamiokande atmospheric data; $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$). Status: STANDARD.
- υ, υ_b: The argument of the tunnelling exponent F, a ratio of an energy to the bulk mass m_Ψ; $\upsilon_\mathrm{b} = \omega_\mathrm{b}/m_\Psi$ that of the particle χ. Unit: dimensionless. Status: DERIVED.
- $F(\upsilon)$: The tunnelling exponent of the width of the particles χ per unit of m_Ψ/H: the integral of the local decay rate q over the region inside the turning point, divided by m_Ψ/H, as a function of the ratio υ of the energy to the bulk mass. Unit: dimensionless. Value: 0.049 for $\upsilon = 0.866$; $(q_\mathrm{b}/\omega_\mathrm{b})^3/3$ for weak binding. Status: DERIVED.
- P_7: The pressure along ordinary space of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(i)}{}_{(i)}$). Unit: GeV^7. Status: DERIVED.
- U_PMNS: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; its entries $(U_\mathrm{PMNS})_{\alpha i}$ connect the flavour state ν_α with the mass state ν_i. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- $\mathcal{L}_\mathrm{fold}$: The Lagrangian of any one term of the fold (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- ρ_7: The energy density of the matter of the fold above its tension in the seven-dimensional normalisation (minus the component $\mathsf{t}^{(4)}{}_{(4)}$: an energy per unit volume of the six other directions of the fold); $V_*\rho_7$ in the four-dimensional normalisation. Unit: GeV^7. Status: DERIVED.
- q_7: The pressure along the extra times of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(r)}{}_{(r)}$). Unit: GeV^7. Status: DERIVED.
- q_χ: The pressure of the particles χ of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED (for radiation and free fields), ASSUMED (for the baryons).
- q: The local decay rate of a profile of dirac16complex along σ at the frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2 \mathrm{e}^{2H|\sigma|}}$ (imaginary beyond the turning point, where the profile oscillates). Unit: GeV. Status: DERIVED.